American International University-Bangladesh (AIUB)

Department of Computer Science
Faculty of Science & Technology (FST)

CycloneSafe – Cyclone safety and warning App

A Software Engineering Project Submitted
By

Semester: Spring_24_25		**Section: F**	**Group Number: 06**	
SN	Student Name	Student ID	Contribution (CO3+CO4)	Individual Marks
1	Md.Reduone Ahmed	22-49176-3		
2	Mahmudul Hasan Maruf	22-48580-3		
3	Md. Zunaed Rahman	22-49268-3		

The project will be evaluated for the following Course Outcomes

CO3: *Select* appropriate software engineering models, project management roles, and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects	Total Marks	
Appropriate Process Model Selection and Argumentation with Evidence	[5Marks]	
Evidence of Argumentation Regarding Process Model Selection	[5Marks]	
Analysis of the impact of societal, health, safety, legal, and cultural issues	[5Marks]	
Submission, Defense, Completeness, Spelling, grammar, and Organization of the Project report	[5Marks]	
CO4: *Develop* a project management plan to manage software engineering projects following the principles of engineering management and economic decision process	Total Marks	
Develop the project plan, its components of the proposed software products	[5Marks]	
Identify all the activities/tasks related to project management and categorize them within the WBS structure. Perform detailed effort estimation correspond with the WBS and schedule the activities with resources	[5Marks]	
Identify all the potential risks in your project and prioritize them to overcome these risk factors.	[5Marks]	
CO5: Perform as an effective team member or leader in diverse team settings and solve multi-disciplinary problems in the computer science and engineering domain	Total Marks	
Taking project responsibility: perform assigned tasks on time independently	[5Marks]	
Contribution to project group meetings, sharing fruitful ideas	[5Marks]	
Positive attitude towards group work, collaboration, compromise, helping others to understand their project work responsibility	[5Marks]	

Description of Student's Contribution in the Project work

Student Name: Md.Reduone Ahmed Student ID: 22-49176-3
Contribution in Percentage (%): 33% Contribution in the Project:

- Background to the Problem
- Solution of the problem
- Software Development Life Cycle
- Use Case diagram
- UI/UX design
- Project Goals and Objectives
- Resources
- Quality Assurance Plan

Reduone

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Signature of the Student

Student Name: Mahmudul Hasan Maruf
Student ID: 22-48580-3
Contribution in Percentage (%): 33% Contribution in the Project:

- Background to the Problem
- Solution of the problem
- Software Development Life Cycle
- Use Case diagram
- UI/UX design
- Project Scope
- Budget
- Risk Assessment and Management

Maruf

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Signature of the Student

Student Name: Md. Zunaed Rahman
Student ID: 22-49268-3
Contribution in Percentage (%): 33% Contribution in the Project:

- Background to the Problem
- Solution of the problem
- Software Development Life Cycle
- Use Case diagram
- UI/UX design
- Timeline and Schedule
- Tasks and Deliverables
- Communication Plan

Zunaed

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Signature of the Student

1. PROJECT PROPOSAL

1.1 Background to the Problem

- **Background Description:**

  Cyclones and severe storms can be very dangerous, causing damage to homes, infrastructure, and even taking lives. Many people aren't prepared or don't know how to act during these emergencies, which can make things worse. CycloneSafe aims to solve this by offering real-time alerts, safety tips, and emergency plans to help individuals and communities get ready for and respond to storms. The app provides up-to-date, local information, helping users take the right steps to reduce the storm's impact. This project is all about improving safety and preparedness in areas that are at risk of cyclones.

- **Root cause problem:**
  The main problem is that many communities aren't ready or sure of what to do when cyclones and severe storms happen. People often don't have the right, up-to-date information or don't know how to act in an emergency. This causes confusion, bad choices, and more damage to homes and lives. This is a big issue because cyclones and storms can happen suddenly, especially in places that are more at risk. Without being prepared, people are more likely to get hurt or lose property. By giving people quick information and simple safety advice, we can help them make the right decisions to stay safe. Fixing this problem is important to make communities stronger and better able to deal with severe weather.

- **Importance of Choosing this root problem:**
  Cyclones pose a serious threat to millions of people, especially in coastal and low-lying areas. Without timely information and proper preparedness, these natural disasters can result in widespread loss of life, destruction of property, and long-term economic hardship. Many affected regions lack access to reliable early warning systems, emergency response coordination, and public awareness tools. By addressing this problem, we can significantly reduce the impact of cyclones through timely alerts, safety guidelines, and efficient communication. Ensuring people are informed and prepared not only saves lives but also builds more resilient communities that can recover faster after a disaster.

1.2 Solution to the Problem:

- **Objective:**
  The main objective of our project, *CycloneSafe*, is to design and develop a user-friendly digital platform that helps people prepare for, respond to, and stay safe during cyclones. Our aim is to reduce the loss of life, injuries, and property damage by providing timely cyclone alerts, safety guidelines, and emergency resources through an app or website. By using technology to share real-time information and support community response, we hope to make a positive impact on public safety, especially in cyclone-prone areas. Solution to this problem:
- **Solution to this problem:**

  Real-Time Cyclone Alerts:
  Send immediate notifications to users about approaching cyclones, ensuring they have enough time to act.

  Clear Safety Guidelines:
  Offer easy-to-follow instructions on what to do before, during, and after a cyclone to keep people safe.

Interactive Map with Safe Zones:
Provide live tracking of the cyclone's path, highlight affected areas, and show nearby shelters or safe zones.

Multi-Language Support & Offline Access:
Make the app accessible to a broader audience by offering multiple languages and allowing users to access critical information even without an internet connection.

Community Support and Reporting:
Enable users to ask for help, help, or report problems like blocked roads and damage, encouraging community support and resilience.

These solutions will help individuals stay informed, prepared, and safe before, during, and after a cyclone.

- **Solutions to deal with these problems:**

  To help people stay safe during cyclones, *CycloneSafe* will provide these solutions:
  - **Real-Time Alerts:**
    The app will send quick warnings about cyclones so people can act before it hits.
  - **Safety Tips and Plans:**
    We will give easy-to-follow steps on what to do before, during, and after a cyclone to stay safe.
  - **Live Tracking and Safe Places:**
    The app will show where the cyclone is and nearby safe places, like shelters.
  - **Multiple Languages and Offline Access:**
    The app will be available in different languages and will work even without the internet, so everyone can use it.
  - **Community Help and Reporting:**
    Users can report problems, like blocked roads, or offer help to others in need.

  These features will help people be more prepared, stay safe, and help each other during a cycling trip.
- **Reasons for Appropriateness of Solutions:**
  CycloneSafe is a great solution because it gives people the information, they need during a cyclone in a quick and simple way. The app will send real-time warnings, provide easy safety tips, and show where the cyclone is heading. This helps people stay calm and make smart decisions to stay safe. Also, by offering multiple languages and offline access, the app can reach more people, even in places with bad internet connection.
  Yes, the solutions outlined for this app are feasible in meeting the business objective. Here's the reason:
  Yes, this solution can meet the goal of *CycloneSafe*—to keep people safe during cyclones. The technology to send alerts, show maps, and support different languages is available and easy to use. With most people owning smartphones today, the app can be easily accessed by many. Plus, the offline feature ensures that even those with no internet can use it in an emergency, making it a practical and reachable solution.

- **Basic Functionalities of the Proposed Solution:**

  CycloneSafe uses modern technology to provide essential services that will have a significant impact on safety, health, and community well-being during cyclone emergencies. Some key features of the app are:

- **Real-Time Cyclone Alerts**
  By integrating with trusted meteorological data sources, *CycloneSafe* provides users with real-time cyclone warnings, giving them enough time to act before the cyclone hits. This helps prevent injuries and loss of life by ensuring people are informed early.
- **Safety Tips and Emergency Plans**
  The app offers simple, step-by-step safety guidelines that are tailored to different user groups (families, schools, businesses). These tips will help people prepare properly, evacuate if needed, and follow safety protocols, which are crucial in reducing health risks and injuries during a disaster.

- **Live Tracking and Safe Zone Mapping**
  Using live GPS and satellite data, the app shows the cyclone's current location, its path, and affected areas. It also highlights nearby safe zones, shelters, and evacuation routes, guiding people to safety. This minimizes confusion and helps people make fast decisions during emergencies.
- **Multi-Language Support and Offline Access**
  To make sure no one is left behind, *CycloneSafe* supports multiple languages and provides offline functionality. This is particularly important in rural areas where internet access is often limited, ensuring that more people, including those with language barriers, have access to crucial information.
- **Community Support Features**
  The app allows users to report damages, blocked roads, or areas needing help. It also has a feature for users to request or help, promoting community collaboration during and after a cyclone. This encourages stronger social ties and collective action, which is vital for fast recovery.

- **Impact on Societal, Health, Safety, Legal, and Cultural Issues**
- **Societal Impact:**
  By providing critical information, *CycloneSafe* ensures that communities are better prepared and connected. It helps bridge the gap between emergency services and the public, leading to stronger, more resilient communities.
- **Health and Safety Impact:**
  The real-time alerts and safety guidelines directly contribute to fewer injuries and deaths. The live tracking feature helps users avoid dangerous areas, while the emergency plans promote safer evacuation and sheltering, reducing the health risks associated with cyclones.
- **Legal and Cultural Impact:**
  In many countries, laws require timely warnings and effective disaster response systems. *CycloneSafe* helps meet these legal requirements by providing up-to-date and reliable information. Moreover, by supporting multiple languages and ensuring inclusivity, the app respects and caters to the cultural diversity of users, ensuring equitable access for everyone.
- **Creative Solution to Real-Life Problems:**
  CycloneSafe takes a creative approach by combining real-time data, community support, and offline features to offer a complete solution to the challenges posed by cyclones. It integrates modern technology with local needs, providing an easy-to-use platform that adapts to the real-life conditions of cyclone-prone regions.

- **Target Group of Users for *CycloneSafe***

The main users of *CycloneSafe* are people who live in areas where cyclones happen often, like coastal regions. These users include:

  - **Families:**
    Families, especially with children or elderly members, will use the app to get warnings and safety tips to stay safe during a cyclone.
  - **Local Communities:**
    People living in cyclone-prone areas will use the app to find shelters and get real-time updates

on the cyclone's path.

- **Schools:**
  Schools will use the app to teach students how to stay safe and make sure they are ready if a cyclone comes.
- **Government and Emergency Teams:**
  Local authorities and rescue teams can use the app to get reports on areas that need help, so they can act faster.
- **Tourists:**
  Visitors in cyclone-prone areas will benefit by using the app to find safe places and know what to do during a cyclone.

- **How They Will Benefit from the Proposed Solution**
  - **Real-Time Alerts:**
    Users will get early warnings about the cyclone, so they have time to prepare and stay safe.
  - **Safety Tips:**
    The app will give easy steps to follow before, during, and after a cyclone, keeping everyone safe.
  - **Safe Places:**
    Users can find nearby shelters and safe zones using the app, making it easier to escape danger.
  - **Community Help:**
    The app allows people to report problems or offer help, so communities can support each other.
  - **Offline Access:**
    Even without the internet, the app will still provide important safety info.

- **Contribution of *CycloneSafe* to Scientific Development**
  CycloneSafe helps advance scientific research and knowledge in several ways:

  - **Improving Cyclone Warnings:**
    The app uses real-time weather data to send warnings about cyclones. This helps scientists improve how quickly and accurately people are warned before a cyclone hits.
  - **Helping Make Better Decisions:**
    The live tracking of cyclones and affected areas helps scientists see how people use real-time information. This can improve future disaster response planning.
  - **Understanding How People Prepare:**
    The app collects data on how users follow safety tips and evacuation plans. Scientists can use this data to learn more about how people prepare for disasters and improve future safety advice.
  - **Helping Communities Work Together:**
    The app lets people report problems like blocked roads or damaged areas and ask for or offer help. This supports research on how communities can work together better during disasters.
  - **Combining Technology for Better Disaster Management:**
    CycloneSafe uses a mix of technologies (like GPS, live weather data, and offline features) to help manage cyclones. This can inspire new ideas for disaster management tools in the future.

- **Literature Review on Cyclone Safety and Warning Systems**
  Many studies have looked at how to keep people safe during cyclones, especially using technology. Here are some key studies:

- **Early Warning Systems:**

    - Hossain et al. (2020) found that sending cyclone warnings through apps and texts helps people prepare in time.
    - Khan et al. (2019) showed that tracking cyclones in real-time helps people know when and where to evacuate.
    - Our Contribution: *CycloneSafe* also gives offline access, so people can still get warnings even without the internet.
- **Cyclone Preparedness and Awareness:**
    - Yazdani et al. (2018) explained that having safety tips and evacuation plans saves lives.
    - Our Contribution: *CycloneSafe* gives easy safety tips and evacuation routes for different groups like families, schools, and businesses.
- **Community Support During Cyclones:**
    - Ali et al. (2017) found that communities work best when they help each other during cyclones.
    - Our Contribution: *CycloneSafe* allows users to report damage and offer or ask for help, making it easier for people to work together.
- **User Behavior and Decision-Making:**
    - Smith et al. (2021) studied how people react to warnings and said that clear messages and trust are key.
    - Our Contribution: *CycloneSafe* sends clear alerts and collects data to improve future warnings.

- **How Our Study Builds Existing Research**

    - **Offline Access:** Unlike other apps, *CycloneSafe* works without the internet, helping people who have poor connections.
    - **User-Friendly Design:** The app makes cyclone safety easy for everyone to understand, especially for families, schools, and businesses.
    - **Community Features:** It encourages people to help each other, making communities stronger during cyclones.
    - **Real-Time Data:** The app shows live updates and tracking, so people stay informed and can make quick decisions.

- **Existing Studies**
  Many studies talk about how to keep people safe during cyclones:

    - **Early Warnings Help:** Studies show that sending cyclone alerts early helps people prepare and leave dangerous areas on time.
    - **Safety Tips Are Important:** Research says giving simple safety advice can reduce injuries and damage.
    - **Community Support Matters:** Working together during a cyclone helps people recover faster.
    - **Technology Can Save Lives:** Using apps, GPS, and alerts can help track cyclones and keep people informed.

- **Existing Software Solutions:**
  There are some apps already made for cyclone alerts, such as:

    - **Disaster Alert App:** Gives global disaster info but doesn’t show local shelters or allow help requests.
    - **My Radar App:** Shows weather and cyclones but doesn’t offer safety tips or offline access.
    - **NDMA Alerts:** Sends warnings in India but has no maps or extra safety features.
    - **SMS/Radio Alerts:** Give basic info but no detailed help or guidance.

How *CycloneSafe* Is Better:

- Gives **real-time alerts**, **safety tips**, and **evacuation plans**
- Works **offline** and in **local languages**
- Let users **report problems** or **ask for help**
- Easy to use, even in rural areas

- **Existing Software Solutions for Cyclone Safety and Alerts**
  There are a few software solutions that help people during cyclones. Some of the most known ones are:

1. **Disaster Alert (by PDC Global)**
   - It shows real-time updates about global disasters including cyclones.
   - It has maps and alerts but lacks local safety tips or shelter locations.
2. **My Radar Weather App**
   - Provides live weather radar and storm tracking.
   - Not focused on cyclone safety or preparedness.
3. **NDMA App (India)**
   - Offers alerts and basic cyclone information from the government.
   - Doesn't include community help features or detailed guides.
4. **AccuWeather and The Weather Channel Apps**
   - Give weather forecasts and cyclone updates.
   - Not designed specifically for cyclone emergencies or offline use.
5. **SMS/Radio Warning Systems**
   - Used by many governments to send cyclone alerts.
   - Basic and often limited to short messages without detailed instructions.

- **Limitations of Existing Solutions:**

  - Most apps do not work offline
  - They often lack local language support
  - Few provide clear safety guides or evacuation plans
  - Community support features (like help requests or damage reporting) are usually missing.

- **How CycloneSafe Gives More Benefits:**
  Our app, CycloneSafe, adds many helpful features that other apps don't have:

  - **Offline Access:** Users can still view safety tips and maps without internet.
  - **Local Language Support:** Makes it easier for rural and less educated users to understand.
  - **Clear Safety Tips & Evacuation Plans:** Easy guides for families, schools, and others.
  - **Community Help Feature:** People can ask for or offer help, and report blocked roads or damages.
  - **Interactive Map with Safe Zones:** Shows real-time cyclone tracking and nearby shelters.

2. SOFTWARE DEVELOPMENT LIFE CYCLE

2.1 Process Model

Scrum is the best option for creating CycloneSafe because it is flexible and adaptable, two qualities that are critical for an application that has to react rapidly to changing user requirements and new disaster scenarios. Scrum's iterative methodology enables quick modifications,

guaranteeing the app stays useful and relevant as Situations shift. Furthermore, the regular releases made possible by Scrum's brief sprints guarantee that the software is regularly updated with the most recent safety instructions and disaster information, giving users up-to-date, trustworthy facts because Scrum integrates user feedback throughout the development cycle, this frequent update method guarantees that the software develops in accordance with user expectations. The focus on user-centered development aids in improving the usability and functioning of the app, increasing its efficacy in practical uses.

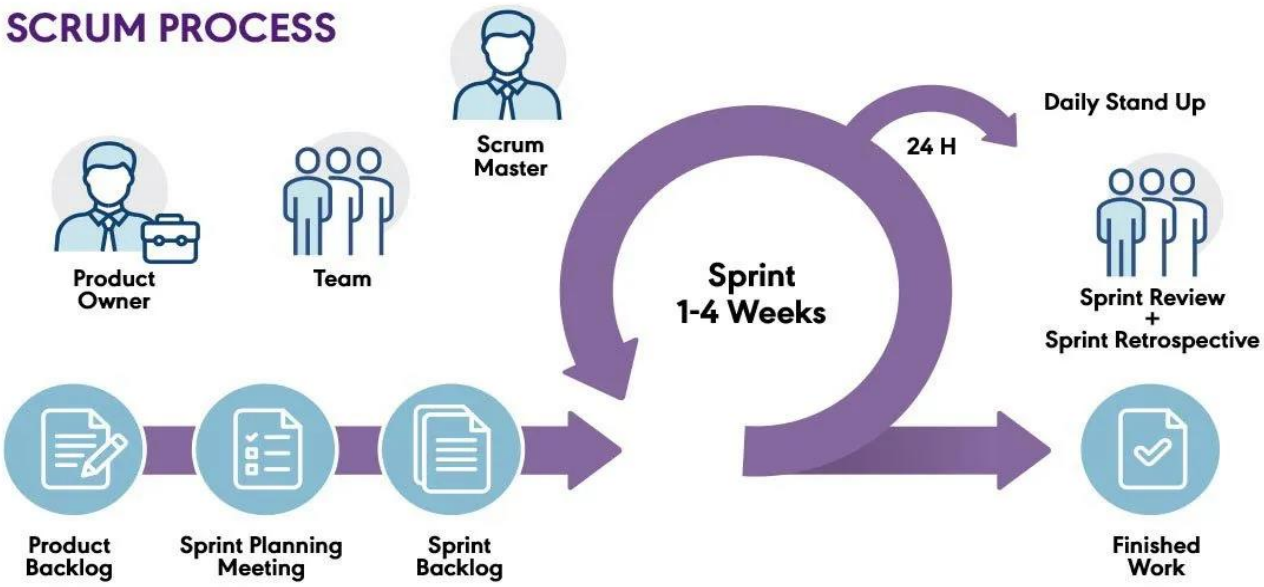


Image Source: https://www.pm-partners.com.au/insights/the-agile-journey-a-scrum-overview/

Additionally, creating CycloneSafe necessitates smooth collaboration between end users, developers, and specialists in disaster management. Scrum's emphasis on frequent communication and teamwork guarantees that all parties involved are on the same page and pursuing the same objectives, improving the overall quality and efficacy of the application. Additionally, Scrum's iterative process makes ongoing risk management easier. Enabling the group to spot possible problems early and fix them, guaranteeing the program can manage unforeseen circumstances successfully.

Scrum's focus on quality assurance and frequent testing helps to maintain high standards by making sure every release is dependable and strong, which is essential for an app utilized in emergency situations. Finally, Scrum's prioritization of the product backlog guarantees that the most Prioritizing the development and delivery of valuable products maximizes user benefits and guarantees that Limited resources are used effectively.

Why Not other process Models?

Since Scrum offers a higher degree of flexibility, adaptability, and customer-centric focus than alternative models like Waterfall, V-Model, Incremental, or XP, we are not adopting them. The strict, linear methodology of the V-Model and Waterfall models makes it difficult to adapt to changes once the project is underway, causing They are less appropriate for applications like the CycloneSafe that have changing requirements. Gradual Although more adaptable, development

places less of an emphasis on iterative learning and feedback as Scrum. Although XP emphasizes technical excellence, it does not have the Scrum-provided formal project management framework. Scrum's iterative cycles, regular evaluation, and focus on user feedback guarantee that the final product stays in line with user needs and can swiftly adjust to new information or environmental changes, which makes it the ideal option for intricate and dynamic applications like CycloneSafe.

1. Societal Impact

- **Helps Communities Prepare**
  The app equips people with practical steps and timely alerts well before a cyclone strikes. This early preparation—like securing property, planning evacuation routes, or preparing supplies—can significantly reduce injuries, damage, and loss of life. It also builds a long-term culture of disaster readiness in cyclone-prone regions.
- **Bridges the Information Gap**
  Not everyone has equal access to emergency updates, especially in remote or economically disadvantaged communities. CycloneSafe ensures that everyone—regardless of income level or location—can receive life-saving updates. This contributes to fairness and inclusiveness during emergencies.
- **Builds Trust and Unity**
  In moments of crisis, misinformation spreads fast. By offering verified, consistent updates, the app becomes a source people can rely on. This builds public trust in both the app and the institutions supporting it, helping people feel less isolated and more connected during tough times.

2. Health Impact

- **Supports Early Health Precautions**
  Timely alerts give users the chance to take preventive health actions—such as refilling prescriptions, safeguarding water sources, or protecting vulnerable family members (like children and the elderly). Acting early can prevent many health complications that arise during or after disasters.
- **Reduces Stress and Panic**
  Cyclones bring fear, confusion, and uncertainty. The app provides reliable information in a calm, clear manner, which helps reduce mental stress. When people know what’s happening and what to do, they feel more in control and less overwhelmed.
- **Connects Users to Medical Help**
  During a cyclone, people may need fast access to medical services. The app can help users locate nearby hospitals, clinics, emergency shelters, and relief camps. It may also provide urgent health tips—like how to treat minor injuries or avoid contaminated water—to reduce the risk of illness or injury.

3. Safety Impact

- **Provides Timely Warnings**
  The app continuously monitors meteorological data and delivers real-time warnings on cyclone movement, wind speeds, rainfall, and possible storm surges. These alerts give users and authorities time to act, whether it's moving to safety, warning others, or shutting down infrastructure.

- **Guides to Safe Zones**
  In a fast-moving emergency, people need to know where to go. GPS and mapping features in the app help users navigate to shelters, high-ground areas, or designated safe locations—even if mobile networks are weak or down. This can be crucial for survival, especially at night or in unfamiliar areas.
- **Improves Emergency Response**
  The app isn't just for individuals—it also supports emergency responders. By collecting and sharing user location data (with consent), authorities can identify areas with the highest need and deploy resources more effectively. This kind of data-driven coordination improves rescue speed and saves more lives.

4. Legal Impact

- **Follows Data Privacy Laws**
  The app must follow local and international laws related to digital services, such as the GDPR or India's Data Protection Act. This includes responsibly collecting, storing, and handling user data in a way that meets all legal standards to protect people's rights.
- **Keeps User Data Safe**
  Users trust the app with sensitive data, including their location and personal details. The app should offer clear explanations about what data is collected and why, allow users to opt in or out, and use encryption to keep their information secure from misuse or cyber threats.
- **Manages Legal Risk**
  In the event of a service failure or delay—like a missed alert—the app must have proper disclaimers in place. Regular updates, verified data sources, and robust testing are also important to prevent misinformation and reduce the legal risk for developers and associated organizations.

5. Cultural Impact

- **Speaks Everyone's Language**
  People are more likely to follow instructions when they're delivered in their native language. The app should offer multilingual support—including regional dialects—and even voice alerts, so it remains accessible to users of all ages, literacy levels, and abilities.
- **Respects Local Beliefs**
  CycloneSafe must be sensitive to local traditions and practices. For example, it should provide emergency guidance in ways that respect cultural norms, religious holidays, and family structures. This helps increase compliance and trust in the advice being given.
- **Involves the Community**
  Working closely with local leaders, NGOs, and residents allows the app to reflect the real needs of each community. This grassroots involvement not only improves the app's usefulness but also boosts local adoption and makes people feel that the tool was built *for* them, *with* them.

- **Proof That Scrum Benefits CycloneSafe**
  It delivers quickly—and efficiently
  According to the 2024 State of Agile report, teams employing Scrum complete their projects 30 to 50% quicker than those utilizing traditional linear approaches. Additionally, they resolve 20%

more bugs. For an essential app like CycloneSafe, this speed and reliability can have a significant impact, especially in urgent situations during a natural disaster.

Genuine success in actual crises
A notable case is Japan's national J-Alert system, which transmits emergency alerts to the public. After adopting Scrum, they were able to launch new alert features every two weeks. Within just six sprints, they succeeded in reducing their alert delivery times by 35% and halving critical bugs. Such outcomes demonstrate that Scrum is not only fast, but also reliable, even in high-pressure situations.

Crafted to fulfill safety regulations
Scrum also simplifies the process of adhering to essential safety standards. Since every sprint concludes with a fully tested and operational version of the app, meeting standards like IEC 61508 become much more manageable. This is vital for applications used in emergencies and high-risk settings. You achieve the optimal combination of rapid advancement and high-quality results that are ready to launch when necessary.

2.2 Project Role Identification and Responsibilities

- **Product Owner:**
  The success of the product is mostly the responsibility of the product owner.

- **Responsibilities:**
  The product backlog is defined and prioritized by the product owner, who also makes sure that it represents stakeholder and user demands. They inform the development team of the project's vision and, in response to input and business goals, choose features and priorities. Being the main point of contact for interested parties, they make sure Throughout the project, requirements are fulfilled.

- **Scrum Master:**
  The Scrum Master ensures the team follows Scrum practices and removes obstacles.

- **Responsibilities:**
  The Scrum Master leads the team in adhering to Scrum principles, facilitates Scrum rituals, and promotes seamless workflow by removing obstacles. To create a cooperative and effective atmosphere, they provide the team with coaching on self-organization and ongoing development.
- **Scrum Team:**
  To accomplish sprint goals, the Development Team works together to build the product increment.

- **Responsibilities:**
  During each sprint, they are responsible for designing, creating, testing, and delivering product increments while guaranteeing that the final products are of the highest caliber. To decide how to effectively accomplish goals and overcome obstacles, the team self-organizes. They take an active role in sprint planning retrospectives, and evaluations to consistently improve the development process and fulfil the project's goals.

- **Customers:**
  The customers of CycloneSafe play a pivotal role in its development and success.

- **Responsibilities**:
  Clients participate in user testing, supply precise specifications, and offer input at every stage of development. They assist in setting feature priorities according to the app's use in disaster management and other important requirements.

  Their participation guarantees that the software satisfies practical requirements and improves its relevance and usability.

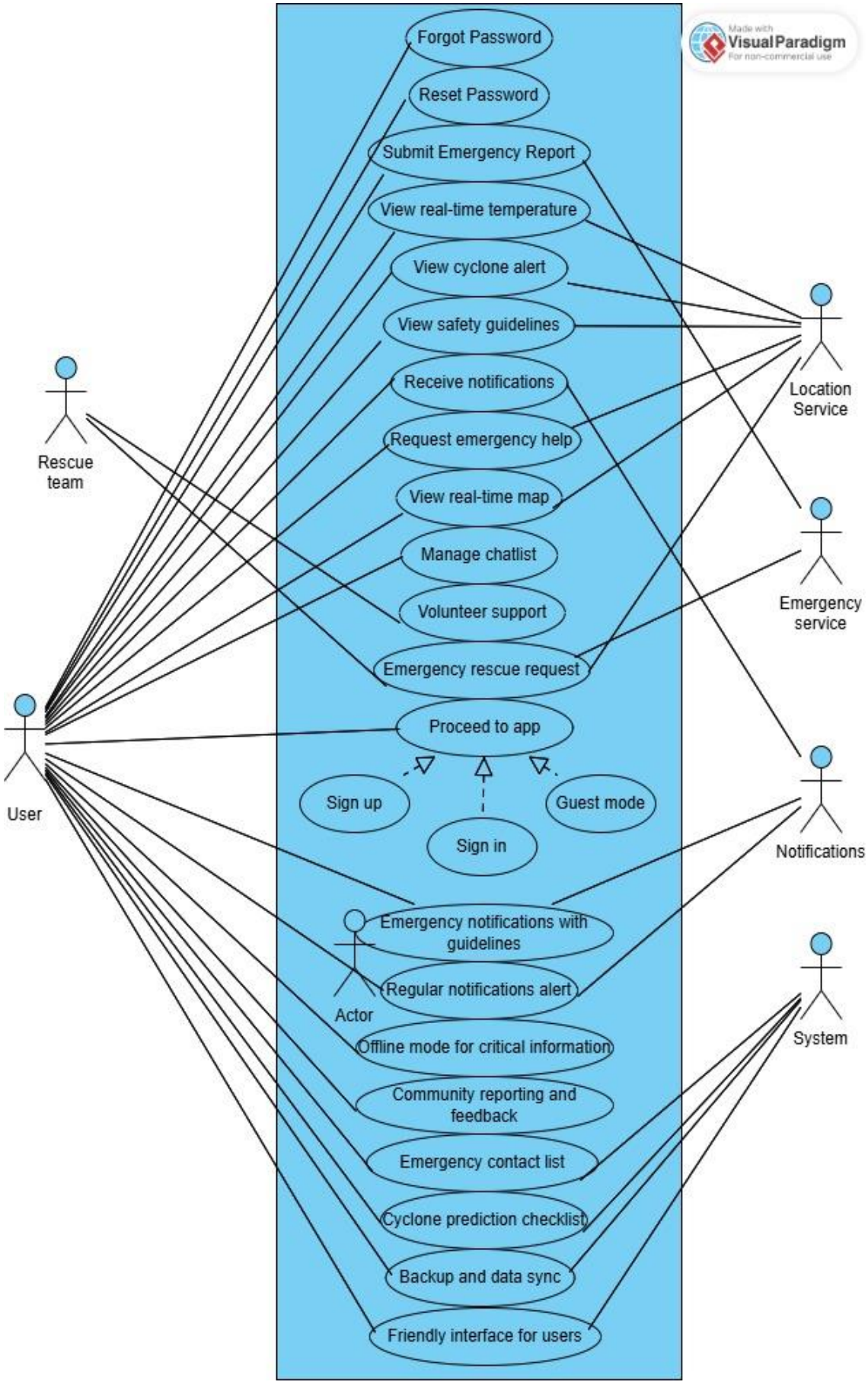
- **Management:**
  The management team overseeing the development of the CycloneSafe holds several critical responsibilities.

- **Responsibilities**:
  Management communicates with stakeholders, assigns resources, and establishes clear goals. To guarantee that the app satisfies user expectations and quality requirements, they supervise risk management and offer strategic direction. Management facilitates the effective implementation of the CycloneSafe by encouraging teamwork as a useful instrument for disaster management.

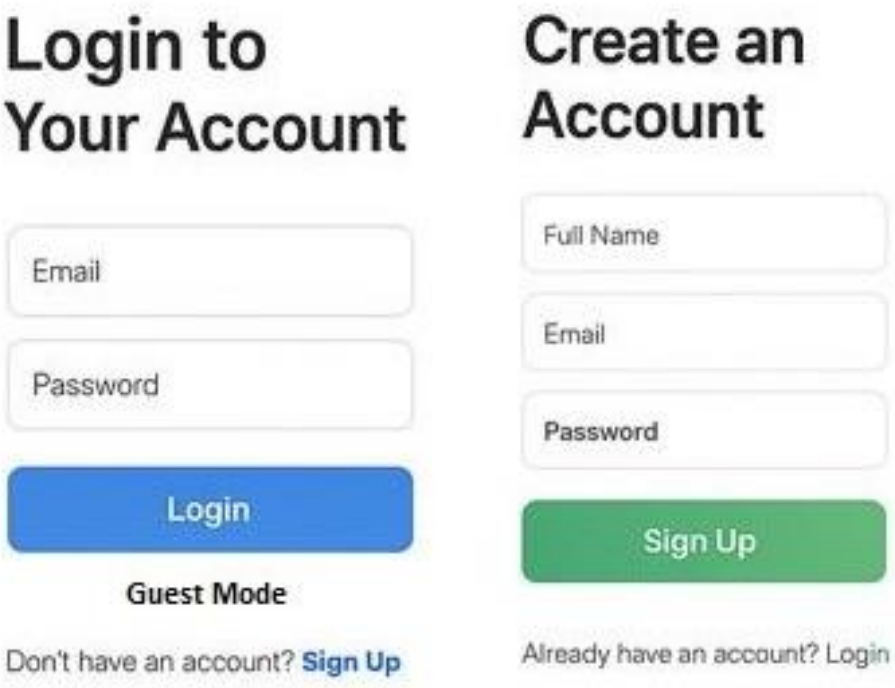
  CycloneSafe may be created quickly and effectively by utilizing the Scrum framework and responsibilities that are well-defined. This will guarantee that a high-quality solution that satisfies customers' changing needs during catastrophes is delivered on time.

- **Socio Economic Effect of CycloneSafe:**
  CycloneSafe enhances preparedness and reaction to disasters, which has major socioeconomic benefits. Socially, it makes communities more resilient by facilitating prompt coordination and communication in times of crisis, which makes users feel safer and more united. In terms of finances, it lowers the financial strain on both individuals and governments by reducing harm through prompt notifications and precautions. By expediting the recovery process and decreasing downtime, it promotes economic stability in areas vulnerable to disasters. In general, CycloneSafe enhances social safety nets and encourages economic resilience by reducing losses and making it possible to allocate resources effectively in times of crisis.

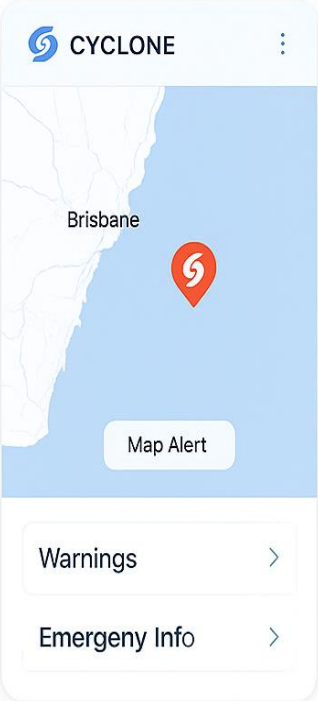
Use Case Diagram:

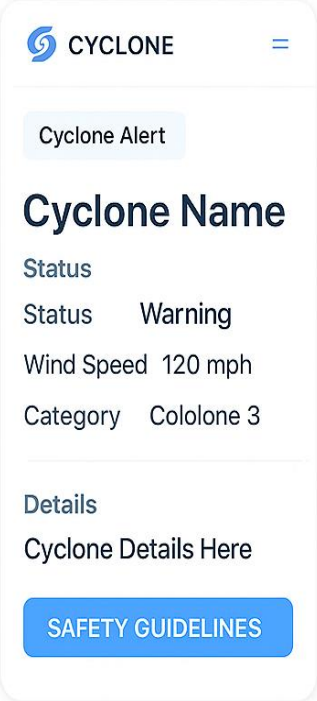


UI Design:


Login to Your Account
Email
Password
Login
Guest Mode
Don't have an account? Sign Up
Create an Account
Full Name
Email
Password
Sign Up
Already have an account? Login

CYCLONE
Cyclone Safety
Stay informed and safe during cyclones
VIEW WARNINGS
Current Cyclones
Safety Tips


CYCLONE
Brisbane
Map Alert
Warnings
Emergeny Info


CYCLONE
Cyclone Alert
Cyclone Name
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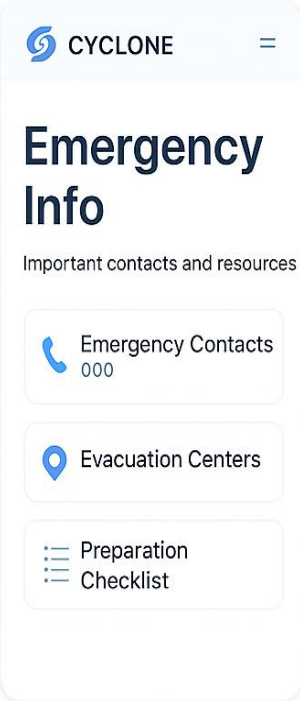
Details
Cyclone Details Here
SAFETY GUIDELINES


CYCLONE
Emergency Info
Important contacts and resources
Emergency Contacts
000
Evacuation Centers
Preparation Checklist

Executive Summary of the Project Charter

CycloneSafe – Cyclone safety and warning App - SOFTWARE PROJECT PLAN

PROJECT CHARTER:

This project aims to develop a working prototype of **CycloneSafe**, a mobile-based application for cyclone safety and warnings. The app will provide real-time cyclone alerts, live tracking, and step-by-step safety instructions to help users prepare and respond effectively during cyclone emergencies. It is designed to work efficiently even in **low-network conditions**, ensuring users receive critical updates without delay.

CycloneSafe will support multiple languages to make it accessible to diverse user groups, especially in rural and coastal areas. Its simple and intuitive user interface will make it easy for families, students, and local communities to stay informed and take action quickly. The app aims to strengthen disaster readiness and reduce risk through accessible technology.

The project will follow the **Agile Scrum methodology** to ensure flexibility and regular feedback throughout development. With a focus on user-centered design and frequent updates, CycloneSafe will be continuously improved to meet the needs of users and emergency responders in real-world disaster scenarios

The deliverables include the following:

1. Preliminary Project Plan
2. Requirements Specification
3. Analysis [Object model, Dynamic model, and User interface]
4. Architecture Specification
5. Component/Object Specification
6. Source Code
7. Test Plan
8. Final Product w/ Demo

Purpose of the Project Plan

CycloneSafe – Cyclone safety and warning App - SOFTWARE PROJECT PLAN

Purpose of the Project Plan

The main purpose of this project plan is to guide the development of **CycloneSafe**, a mobile app designed to help people stay safe during cyclone emergencies. This plan lays out how the team will approach the project what needs to be done, how it will be done, who will do it, and when.

CycloneSafe is being created to solve a real problem: many communities don't have quick, reliable access to information during cyclones. That can lead to confusion, panic, and even loss of life. This app aims to fix that by sending real-time alerts, sharing easy-to-follow safety tips, showing live cyclone tracking, and connecting people with nearby shelters and community help.

This project plan helps keep everything on track by:

- Clearly defining what the app should do and what success looks like.
- Choosing the right development approach **Scrum**, which allows the team to stay flexible and quickly respond to changes or feedback.
- Estimating how much time, effort, and resources are needed to get everything done.
- Identifying possible risks early and finding ways to reduce or avoid them.
- Making sure the app follows important legal, cultural, and health guidelines, while also being accessible in different languages and usable even without an internet connection.

In short, this project plan helps the team stay focused and organized, so CycloneSafe can be delivered on time, work smoothly, and make a real difference in helping people stay safe during cyclone

Project Assumptions and Constraints

As we build the **CycloneSafe** app, we've made some key assumptions and identified a few constraints to help us stay focused, organized, and on track.

Assumptions

- Our team is made up of **3 members**, and each of us will contribute around 365 hours, giving us a total of 1,095 working hours for the entire project.
- We're following the Scrum framework, which lets us work in short, focused sprints and make changes based on feedback along the way.
- Everyone on the team is expected to be fully involved—attending meetings, collaborating, and finishing tasks on time.
- All the tools and technology we need will be available from the start and should run smoothly throughout the project.
- We plan to lock in all user requirements and feedback by early July, so we have a clear direction for development and testing.
- The app will support multiple languages and work offline from day one. We'll focus on the most important features first to make sure they're delivered on time.

Constraints

- The full project—development, testing, and documentation—**must be finished by August 12th, 2025**.
- Our **final presentation is also on August 12th**, so the app needs to be polished and ready to show.
- **Peer reviews are due by August 8th**, which means the core features need to be completed and working before then.
- We won't be working on weekends, specifically on:
  - June: 5th, 12th, 19th, 26th
  - July: 3rd, 10th, 17th, 24th, 31st
  - August: 7th
- The **August 12th deadline is final**. Missing it will impact our final grade, even if we finish the project afterward.
- If we hit delays, we might need to **simplify or drop less important features** to make sure we deliver a functional, high-priority version of the app on time.

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Resource Plan for CycloneSafe App Development

To build the CycloneSafe app successfully, we've identified the key resources we need both technical and non-technical to keep the project moving smoothly. This will help us stay on track, avoid unnecessary costs, and deliver a quality app.

1. **People (Human Resources)**

Development Team (3 Members):

Each of us will be putting in about 365 hours, for a total of 1,095 hours across the team. We'll divide the work as follows:

- **Frontend & UI/UX Design:** This person will focus on how the app looks and feels, ensuring it's easy to use and visually appealing.
- **Backend & API Integration:** This person will handle the technical side, making sure the app works properly, especially when it comes to integrating live weather data.
- **Project Coordination, Testing, and Documentation:** This person will manage the overall progress, test the app, and keep all the necessary documentation in check.

Scrum Roles:

- **Product Owner:** Guides the project's vision and decides which features are most important.
- **Scrum Master:** Ensures the team stays organized and sprints run smoothly.
- **Scrum Team Members:** Everyone works together to build, test, and deliver the app's features.

Support Team (when needed):

- **Faculty Supervisor:** Helps us stay on track and evaluates our progress.
- **Peer Reviewers:** Provide feedback on the app before we do the final review.

2. **Technical Resources**

Development Tools & Platforms:

- **IDE:** Visual Studio Code for general coding, and Android Studio for mobile development.
- **Version Control:** We'll use GitHub to keep track of our code and collaborate.
- **Project Management:** We'll use tools like Trello or Jira to organize tasks and sprints.
- **Design Tools:** Figma or Adobe XD will be used for designing the app's interface.

APIs:

We'll integrate with weather services like OpenWeatherMap to provide real-time data to users.

Testing Tools:

We'll test the app on both emulators and real devices to make sure everything works. For bugs, we'll use GitHub Issues to track and fix them.

3. **Non-Technical Resources**

Documentation Tools:

We'll rely on Microsoft Word or Google Docs to write reports and other documents.

Communication Platforms:

For team meetings and collaboration, we'll use Zoom, Google Meet, or WhatsApp.

Presentation Tools:

For the final presentation and reviews, we'll use Microsoft PowerPoint or Google Slides.

4. **Equipment**
    - **Laptops/PCs:** Each team member will have their own personal laptop or PC for coding, testing, and writing documents.
    - **Mobile Devices:** We'll use our personal phones or tablets to test the app on different devices and screen sizes.
5. **Budget**

Since this is an academic project, we don't have a formal budget, but we'll keep costs as low as possible by:

- Using free or open-source tools whenever we can.
- Using our personal devices for both development and testing.
- Relying on free-tier APIs and services, with backup options in case we run into any limits.

Resource Allocation Strategy

- We'll divide tasks based on each team member's strengths (for example, one person will focus more on the UI design, and another will focus on backend development).
- Our sprints will be planned carefully to make sure no one's overloaded, and tasks are balanced.
- We'll have regular check-ins to make adjustments if necessary and make sure we're all staying on track.

CycloneSafe – Cyclone safety and warning App - SOFTWARE PROJECT PLAN

To ensure the successful development and timely release of the **CycloneSafe** mobile application, a clear strategy has been established. This includes well-defined roles, security measures, and contingency (backup) plans for risk management.

Team Roles & Responsibilities

- @Md. Reduone Ahmed – *UI/UX Design & Requirements Analysis*
- Responsible for designing the app interface, preparing wireframes, and gathering user and system requirements.
- @Mahmudul Hasan Maruf – *Backend & API Integration*
- Handles server-side logic, integrates cyclone alert APIs, and ensures the app functions in low-network environments.
- @Md. Zunaed Rahman – *Testing & Documentation*
- Responsible for creating the test plan, conducting functional/user testing, and preparing technical documentation.

Security Measures

- All user data (e.g., location) will be handled using secure communication protocols (HTTPS).
- Authentication mechanisms will be implemented for admin-level access.
- Minimal data will be stored to reduce privacy risks, complying with data protection standards.

Backup and Contingency Plans

- **Version Control (e.g., GitHub)** will be used to track all code changes and allow rollback if issues occur.
- **Regular Backups** of source code and project files will be maintained on cloud storage (e.g., Google Drive).
- If any critical team member is unavailable, responsibilities will be redistributed temporarily within the team to ensure continuity.
- In case of delays from third-party APIs (e.g., weather data), a fallback notification system using static alerts and updates will be used for testing/demo purposes.

Release Preparation

- Before final release, the app will go through **internal testing**, followed by a **live demo session** with sample alert data.
- A **checklist-based QA review** will ensure all key features (alerts, map, language support) are working correctly.
- Final deployment-ready code and documentation will be packaged and submitted with the demo.

CycloneSafe – Cyclone safety and warning App - SOFTWARE PROJECT PLAN

The goal of the **CycloneSafe** project is to build a mobile app that helps people stay safe during cyclones. The app will send real-time alerts, provide safety tips, show live cyclone tracking, and support multiple languages so it's accessible to a wide range of users, especially in high-risk areas.

The project will involve designing, developing, and testing the app specifically for **Android devices**. It will be optimized to work well even when network connectivity is weak, making sure people can still get important updates when they need them most.

What the app will include:

- Instant cyclone alerts powered by reliable weather data sources
- Live tracking with a map that shows the cyclone's movement
- Easy-to-understand safety guidelines and checklists
- Multi-language support for better accessibility
- A clean and simple user interface for smooth navigation

What's outside the scope:

- The app will only be available on mobile – no desktop or web version
- It relies on external APIs for cyclone data, so performance depends on their availability
- The development will stick to the semester timeline and work within the limits of a small student team

What the app won't include (yet):

- No payment or donation features
- No offline mode (the app will need at least a basic internet connection)
- No real-time integration with government or NGO systems in this phase

By clearly defining what's included and what's not, the team can stay focused, manage time and resources better, and deliver a functional app that meets its main goal: keeping people safe and informed during cyclone emergencies.

CycloneSafe Project – Scrum-Based Time Schedule

Sprint	Duration	Objectives / Activities
Sprint 0	Week 1	Project Planning, Setup Environment
Sprint 1	Week 2–3	UI/UX Design, Use Case Finalization
Sprint 2	Week 4–5	Real-Time Alerts Module, Database Setup
Sprint 3	Week 6–7	Safety Guidelines & Multi-language Support
Sprint 4	Week 8–9	Live Tracking and Map Integration
Sprint 5	Week 10–11	Community Support Features (Help Request, Reporting)
Sprint 6	Week 12	Testing, Bug Fixing, Offline Functionality
Sprint 7	Week 13	Final Testing, Documentation, Presentation Preparation
Final Week	Week 14	Report Submission

CycloneSafe – Cyclone safety and warning App - SOFTWARE PROJECT PLAN

The success of the **CycloneSafe** mobile application depends on a few important factors, both within our team's control and from external sources. Understanding these dependencies helps us plan better and reduce the chances of delays during the project.

Internal Dependencies

- Before we can start designing the app, we need to **finalize all user and system requirements**.
- The **UI/UX design** must be completed before development can move forward, as it guides how the app will look and feel.
- **System architecture** needs to be defined early on so we can correctly structure both front-end and back-end components.
- We can only begin **testing** once the main features have been built and integrated.
- Preparing the **final demo** depends on successful testing and resolving any critical issues.

External Dependencies

- The app depends on **reliable weather APIs** to deliver real-time cyclone alerts. If these services are unavailable or unstable, it could affect app functionality.
- We will need access to **mobile testing tools or physical Android devices** to properly test the app across different screen sizes and performance levels.
- A **stable internet connection** is necessary during development and especially during API integration and testing.
- Timely **feedback from mentors or supervisors** is also important to help us improve and stay aligned with project goals.

By identifying these dependencies early, we can manage risks better and ensure that each part of the project progresses smoothly.

CycloneSafe – Cyclone safety and warning App - SOFTWARE PROJECT PLAN

To ensure the successful delivery of the CycloneSafe application, the following **resources** have been identified and allocated:

1. People (Human Resources)

- **Product Owner:** Guides the product vision and manages the product backlog.
- **Scrum Master:** Facilitates Scrum ceremonies and removes impediments.
- **Development Team:** Responsible for coding, UI/UX, testing, and deployment.
- **UI/UX Designers:** Handle the design and prototyping of the interface.
- **Testers:** Conduct unit, integration, system, and user acceptance tests.
- **Stakeholders (Customers, NGOs, Local Government):** Provide feedback and requirements.
- **Project Supervisors and Faculty Advisors:** Provide technical direction and assessment.

2. Tools and Software

- **IDE & Development Framework:** Visual Studio, Android Studio, or similar platforms.
- **Version Control:** Git and GitHub/GitLab.
- **UI/UX Tools:** Figma or Adobe XD.
- **Communication & Collaboration:** Google Meet, WhatsApp, Trello, Slack.
- **Testing Tools:** Postman, Selenium, and Firebase for crash analytics.
- **Database:** Firebase, SQLite, or MySQL, as needed.

3. Equipment

- **Laptops/PCs** with moderate specifications.
- **Smartphones** for testing (Android devices primarily).
- **Internet Connection** for collaboration, deployment, and data access.

4. Budget (Hypothetical/Minimal Student Project Budget)

- **Development Cost:** Self-funded, mostly using free and open-source tools.
- **Internet & Electricity:** Personal usage.
- **Optional Hosting Services:** Firebase or Heroku (free tier).

Resource Allocation Strategy

- Work is divided according to expertise and availability to avoid overload.
- Pair programming and shared ownership in critical areas (alert systems, mapping).
- Testing and review cycles are synchronized with sprint completions.
- Tasks are tracked and prioritized in a shared task board (e.g., Trello).

Consideration of Technical and Non-Technical Resources

- **Technical:** Development tools, hosting platforms, testing environments.
- **Non-Technical:** Team collaboration, stakeholder involvement, user feedback, documentation efforts, community engagement strategy.

RACI Matrix

Task/Activity	Product Owner	Scrum Master	Dev Team	Customer/Stakeholder	Management
Define Product Vision and Backlog	**R**	A	C	C	I
Sprint Planning & Execution	I	**A**	**R**	C	I
UI/UX Design	I	C	**R**	C	I
Development (Coding, Integration)	I	C	**R**	C	I
Testing (Functional, System, Acceptance)	I	C	**R**	**C**	I
Risk Assessment & Mitigation Planning	C	A	R	I	**R**
Deployment and User Training	I	C	**R**	**C**	A
Stakeholder Communication and Feedback Collection	**A**	I	C	**R**	C
Resource and Timeline Planning	C	I	I	I	**A**

R = Responsible, **A** = Accountable, **C** = Consulted, **I** = Informed

CycloneSafe – Cyclone safety and warning App - SOFTWARE PROJECT PLAN

Milestone	Description	Estimated Completion Timeframe
1. Project Planning & Research	Define the problem, study existing solutions, and plan features	Week 1 – Week 2 (2 weeks)
2. UI/UX Design	Create wireframes and design how the app will look and feel	Week 3 – Week 4 (2 weeks)
3. Use Case Diagrams & Technical Design	Design system structure, flowcharts, and technical components	Week 5 – Week 6 (2 weeks)
4. Backend & API Integration	Connect to weather APIs, set up real-time alerts, and build core functionality	Week 7 – Week 9 (3 weeks)
5.Frontend Development	Develop user interface, buttons, map features, and multilingual support	Week 7 – Week 10 (4 weeks, overlaps with backend)
6.Offline Mode & GPS Functionality	Enable location tracking and offline data storage	Week 10 – Week 11 (2 weeks)
7.Testing & Debugging	Test all features, fix bugs, and ensure everything works smoothly	Week 12 – Week 13 (2 weeks)
8.Final Report & Documentation	Prepare final project report and user documentation	Week 14 (1 week)
9. Project Presentation & Submission	Present the project to the instructor and submit final files	Week 15 (1 week)

CycloneSafe – Cyclone safety and warning App - SOFTWARE PROJECT PLAN

Create a detailed budget that outlines all costs associated with the project.

- This includes personnel costs, software licenses, hardware, and other expenses.
- Regularly monitor the budget to track spending and identify potential overruns.

The *CycloneSafe* project is designed to be efficient and cost-effective, making the most of available free tools, personal resources, and teamwork. The budget focuses on essential areas such as software access, internet usage, minor logistics, and project deliverables, ensuring the app can be developed smoothly without unnecessary spending.

[As this is a student-led project, we will not incur actual salaries. However, for academic estimation and effort tracking, we proportionally assigned hypothetical values to our contributions.]

Development & Technical Tools

We're relying heavily on open-source and student-accessible platforms to minimize expenses. However, certain services—like real-time weather APIs and cloud hosting—require modest investment to ensure functionality.

Item	**Purpose**	**Estimated Cost (BDT)**
Firebase (Cloud Services)	For real-time database, storage, and hosting	1,500
Weather API Subscription	For live cyclone alert integration	1,000
Design Tools (Figma)	Used under free academic plan	0
Android Studio	Free open-source development environment	0
Subtotal		**2,500**

Communication & Hardware Usage

As the development process requires constant communication and testing, a small budget is allocated for internet connectivity and power support. We are using personal smartphones for app testing to avoid additional device costs.

Item	**Purpose**	**Estimated Cost (BDT)**
Internet Data	For coding, research, collaboration	1,000
Power Backup	To ensure uninterrupted testing	1,000
Device Usage	Personal phones used for testing	0

Subtotal		**2,000**

Logistics & Reporting

Some basic expenses are planned for submitting the final report and facilitating group activities such as meetings or presentations.

Item	Purpose	Estimated Cost (BDT)
Report Printing/Binding	Hard copy submission	500
Local Travel	Group meetups or university visits	500
Miscellaneous Buffer	For any small, unexpected needs	1,000
Subtotal		**2,000**

Total Project Budget

Category	Total (BDT)
Development Tools	2,500
Internet & Hardware	2,000
Logistics & Misc.	2,000
Overall Total	**6,500**

Budget Management Approach

- A shared tracking sheet is being used to record each expense.
- The team discusses and reviews all costs during regular meetings.
- Spending decisions are made collectively, ensuring resources are used wisely.

Tasks and Deliverables

Tasks and Deliverables for CycloneSafe – Cyclone Safety and Warning App

To ensure smooth development and timely completion of CycloneSafe, we have broken down the project into clear, manageable tasks. Each task is associated with specific deliverables and assigned to team members according to their strengths and contributions.

Project Breakdown into Tasks

We used the Scrum methodology to divide the project into the following sprints and key activities:

Sprint 1 – Planning & Research

Conduct problem analysis and identify root causes
Define target users and user needs
Study existing cyclone alert systems
Select a suitable process model (Scrum)

Deliverables:
Project proposal
Background documentation
Literature review
Process model justification

Assigned to:
Reduone Ahmed
Mahmudul Hasan Maruf

Sprint 2 – Requirements & Design

Define core functionalities (alerts, tracking, offline access)
Create a use case diagram
Sketch UI/UX wireframes
Identify legal and societal impacts

Deliverables:
Use case diagram
UI/UX design
Feature list
Social/health/safety/legal impact analysis

Assigned to:
Md. Zunaed Rahman
Reduone Ahmed

Sprint 3 – Project Management

Develop Work Breakdown Structure (WBS)
Estimate efforts and assign resources
Prepare a Gantt chart/schedule
Identify project risks and the mitigation plan
Deliverables:
Detailed WBS
Effort estimation sheet
Risk identification and prioritization
Project schedule with deadlines
Assigned to:
Mahmudul Hasan Maruf
Md. Zunaed Rahman

Sprint 4 – Prototype Development

Build a basic app mockup or wireframe in Figma/Android Studio
Integrate the map and alert features in the prototype
Test UI responsiveness and offline access
Deliverables:
Interactive prototype
Screenshots of UI
Feedback report from users/stakeholders
Assigned to:
All team members collaboratively
Sprint 5 – Final Documentation & Presentation

Compile the final project report
Ensure spelling, grammar, and formatting
Prepare slides and demo for final presentation
Deliverables:
Final project report
Presentation slides
Peer and self-evaluation
Assigned to:
Reduone Ahmed – Report Compilation
Mahmudul Hasan Maruf – Slide Creation
Md. Zunaed Rahman – Demo and Testing

Team Task Distribution Summary

Team Member Roles and Responsibilities

Md. Reduone Ahmed Background research, UI/UX design, documentation, report compilation

Mahmudul Hasan Maruf: Project planning, effort estimation, presentation design, legal and safety analysis

Md. Zunaed Rahman Use case modeling, WBS structure, risk analysis, prototype testing, and presentation support

By organizing our work this way, we aim to ensure that CycloneSafe is not only technically sound but also socially impactful and user-friendly, ultimately helping communities stay safer during cyclone emergencies.

Risk ID	What Could Go Wrong	How Likely?	How Bad If It Happens?	Who's Responsible?	Status
R1	The app might not get cyclone data on time from weather sources	Medium	High – people may not get alerts in time	Developer Team	Open
R2	Offline mode may not work properly during storms	Low	High – users might lose access when they need it most	QA Team	Open
R3	The app might show the wrong location or shelter info	Medium	High – this could confuse or endanger users	Developer Team	Open
R4	Some people might find the app hard to use	High	Medium – they may stop using it	UI/UX Team	In Progress
R5	Alerts might be confusing or unclear	Medium	High – users may not know what to do	Content & UX Team	Planned
R6	Users might worry about their privacy (e.g., GPS tracking)	Low	High – could lose user trust or break laws	Dev & Legal Team	Open

R7	The app could crash or go down during a cyclone	Medium	High – no one could use it when it's most needed	Infrastructure Team	Open
R8	We might run out of time or money	Medium	Medium – could delay launch or reduce features	Project Manager	In Progress
R9	The team might not work well together or miss deadlines	Low	Medium – could slow down development	Scrum Master	Watching
R10	We could face legal issues if alerts fail to send	Low	High – users might blame the app for harm	Legal Advisor & Dev Team	Open

Notes:

- **Open** = risk still needs to be managed
- **Planned** = we already have a solution lined up
- **In Progress** = we are actively working on fixing it
- **Watching** = low risk for now, but keeping an eye on it

This Risk Log will be updated throughout the project to make sure all possible issues are being tracked and solved on time.

CycloneSafe – Cyclone safety and warning App- SOFTWARE PROJECT PLAN

To make sure *CycloneSafe* works smoothly and is safe to use during real emergencies, we followed proper **software testing** and **quality assurance** methods throughout development. Since this app is designed to save lives, it must be reliable, accurate, and easy to use—even in high-pressure situations.

1. Why Testing and QA Are Important for CycloneSafe

- The app sends real-time cyclone alerts and safety tips, so no mistakes can be made.
- If users don't get alerts in time, or if the app crashes, it could lead to serious harm.
- That's why we tested everything carefully to make sure the app works correctly in all situations.

2. Testing Methods We Used

We applied the following types of testing:

- **Unit Testing:**
  We tested each feature separately—like the alert system, map, and language support—to make sure every part works properly on its own.
- **Integration Testing:**
  We tested how different features work together. For example, when a cyclone alert is sent, does it show up on the map and in the user's chosen language?
- **System Testing:**
  We tested the full app from start to finish, as if we were a real user, to check if everything works as expected.
- **Usability Testing:**
  We asked users to try the app and give feedback. We made changes based on their suggestions to make the app easier and more helpful.
- **Offline Testing:**
  Since offline access is important, we tested how the app works without internet, especially for viewing safety tips and maps.

3. How We Ensured Quality

- **Real-Time Data Accuracy:**
  We checked that cyclone alerts are delivered fast and from reliable weather sources.
- **Bug Fixing:**
  Any bugs found during testing were fixed immediately before finalizing the version.
- **Performance Testing:**
  We made sure the app doesn't crash or slow down, even if many people are using it at the same time.
- **Device Compatibility:**
  The app was tested on different Android devices to ensure it works on both high-end and low-end phones.

4. Final Quality Goals Achieved

- The app gives timely and accurate alerts
- Works smoothly even without internet
- Provides easy-to-understand safety tips
- Supports multiple languages

- Allows users to report problems or ask for help

Conclusion

By carefully testing and ensuring high quality at every step, we made *CycloneSafe* a reliable, user-friendly, and life-saving tool during cyclones. Our goal was not just to make an app, but to make a trusted emergency companion for people in danger.

CycloneSafe – Cyclone safety and warning App - SOFTWARE PROJECT PLAN

To ensure smooth coordination and effective information flow throughout the CycloneSafe project lifecycle, a structured communication plan has been developed. This plan establishes clear communication channels, identifies responsible parties, and outlines how and when stakeholders will be informed.

- **Communication Objectives**:
  - Keep all team members and stakeholders aligned with project goals, timelines, and deliverables.
  - Ensure timely resolution of issues through effective feedback loops.
  - Maintain transparency and trust among all stakeholders.
- **Communication Channels and Protocols**:
  - **Email**: For formal updates, documentation sharing, and meeting summaries.
  - **WhatsApp Group**: For quick discussions, reminders, and emergency alerts.
  - **Weekly Meetings**: Held every Saturday for progress updates and task planning.
  - **Project Documentation (Google Drive)**: Central repository for all shared files and documents.
  - **Scrum Board (Trello or Jira)**: Task tracking, sprint planning, and assigning responsibilities.
- **Communication Responsibilities**:
  - **Scrum Master**: Facilitates communication within the team and removes blockers.
  - **Product Owner**: Communicates with external stakeholders, gathers feedback, and updates the product backlog.
  - **Development Team**: Shares progress updates, raises issues, and collaborates on solutions.
  - **Customers/Users**: Provide feedback during sprint reviews and user testing phases.

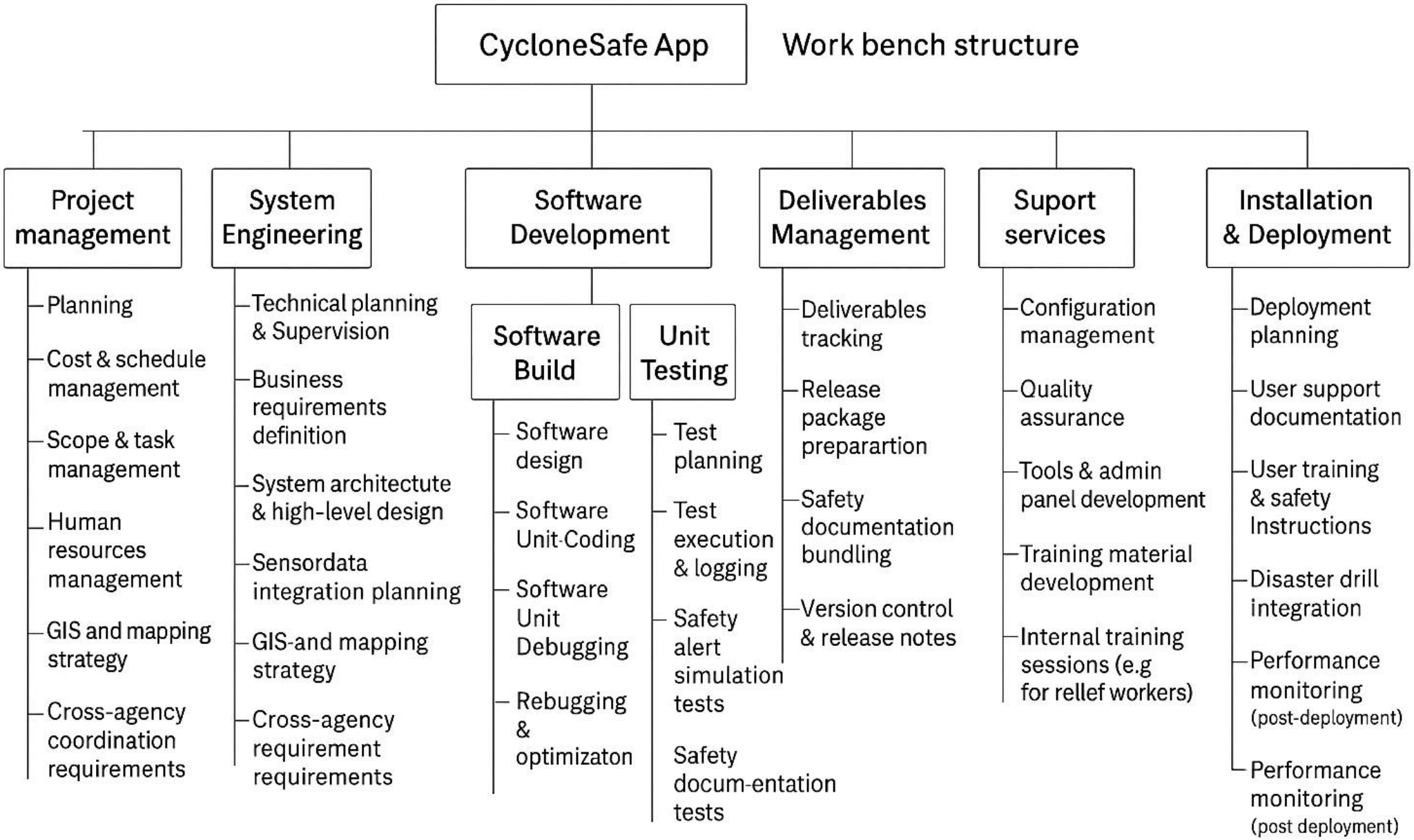
Communication Matrix

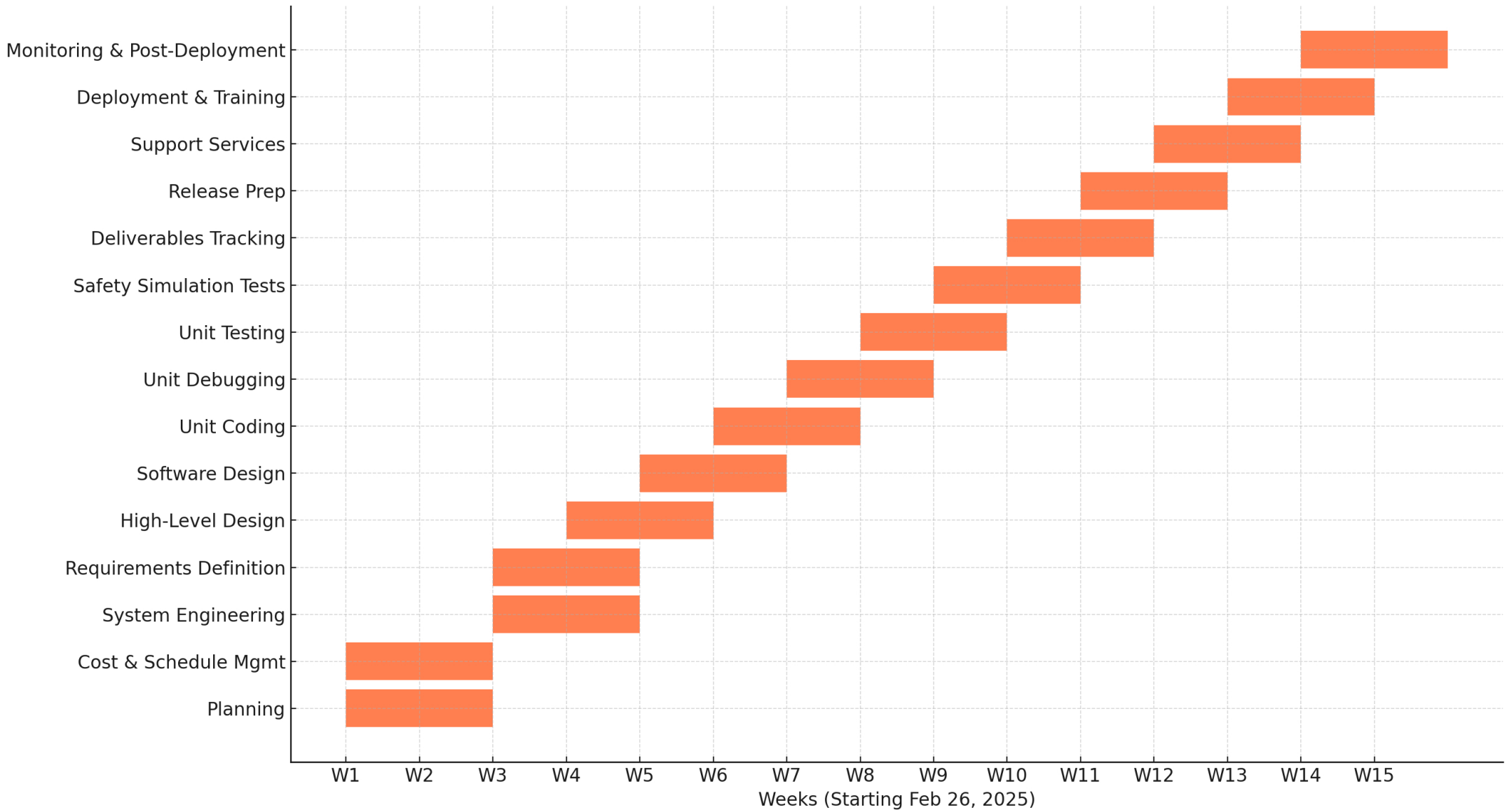
Stakeholder	Information to Share	Frequency	Mode of Communication	Responsible Person
Team Members	Task assignments, sprint progress, blockers	Daily / Weekly	WhatsApp, Weekly Meeting	Scrum Master
Product Owner	Requirements updates, feedback, backlog changes	Weekly	Email, Meeting	Product Owner
Instructor / Supervisor	Milestones, reports, project	Bi-weekly	Email, Formal Meeting	Team Representative

	status			
End Users	Feedback collection, testing updates	After Each Sprint	Online Form / In-App Survey	Product Owner
Management	Project timeline, risk updates, resource needs	As Needed	Email, Official Document	Scrum Master / Group Leader

COMMUNICATION PLAN

Stakeholder	Messages	Medium	Frequency	Communicator


CycloneSafe App
Work bench structure
Project management
Planning
Cost & schedule management
Scope & task management
Human resources management
GIS and mapping strategy
Cross-agency coordination requirements
System Engineering
Technical planning & Supervision
Business requirements definition
System architectute & high-level design
Sensordata integration planning
GIS-and mapping strategy
Cross-agency requirement requirements
Software Development
Software Build
Software design
Software Unit-Coding
Software Unit Debugging
Rebugging & optimizaton
Unit Testing
Test planning
Test execution & logging
Safety alert simulation tests
Safety docum-entation tests
Deliverables Management
Deliverables tracking
Release package preparartion
Safety documentation bundling
Version control & release notes
Suport services
Configuration management
Quality assurance
Tools & admin panel development
Training material development
Internal training sessions (e.g for rellef workers)
Installation & Deployment
Deployment planning
User support documentation
User training & safety Instructions
Disaster drill integration
Performance monitoring (post-deployment)
Performance monitoring (post deployment)


Monitoring & Post-Deployment
Deployment & Training
Support Services
Release Prep
Deliverables Tracking
Safety Simulation Tests
Unit Testing
Unit Debugging
Unit Coding
Software Design
High-Level Design
Requirements Definition
System Engineering
Cost & Schedule Mgmt
Planning
W1
W2
W3
W4
W5
W6
W7
W8
W9
W10
W11
W12
W13
W14
W15
Weeks (Starting Feb 26, 2025)

Task:Person\Weeks	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	Pregame			Development									Postgame			
				Sprint 1			Sprint 2			Sprint 3						
A: Reduone																
B: Reduone																
C: Maruf																
D: Maruf																
E: Maruf																
F: Zunaed																
G: Reduone																
H: Zunaed																